

AuraSense — Architecture & Feature Plan (Updated)

Document purpose: detailed system architecture and practical feature plan incorporating all new ideas and user-specified functionality for the Pfizer & Aava Predictive Migraine Detection challenge.

1. High-level architecture (Mermaid)

```
flowchart TD
    subgraph Mobile
        MApp["Mobile App\n(React Native)"]
        MHK["HealthKit / Health Connect"]
        MCal["Calendar API"]
        MNotif["Notifications & Relax Mode"]
    end

    subgraph Backend
        API["FastAPI Backend"]
        Auth["Auth (Firebase / OAuth)"]
        Ingest["Ingestion Queue (Redis/Celery)"]
        ModelSrv["Prediction Model Service"]
        Features["Feature Store (Firestore / PostgreSQL)"]
        Explain["Explainability (SHAP)"]
        Notif["Notification Service (APNS / FCM)"]
    end

    subgraph External
        Weather["(OpenWeatherMap / Tomorrow.io)"]
    end

    MApp -->|Health, calendar, user inputs| API
    MApp -->|Feedback & migraine logs| API
    MApp --> MNotif
    API --> Auth
    API --> Ingest
    Ingest --> ModelSrv
    ModelSrv --> Features
    ModelSrv --> Explain
    ModelSrv --> API
    API --> Notif
    API -->|weather request| Weather
    Weather --> API
```

2. Core user features

✓ Data collection & analysis

- **Health Data Integration**
 - iOS → Apple **HealthKit**
 - Android → **Health Connect**
 - Collect metrics: sleep hours, HR, HRV, step count, activity duration.
- **Calendar Analysis**
 - Fetch event frequency, duration, and overlaps → compute stress proxy.
- **Weather Analysis**
 - Fetch air pressure, humidity, and temperature via **OpenWeatherMap** → key trigger features.
- **User Input Requests**
 - Daily quick survey: "Did you drink coffee?", "Did you skip a meal?", "Did you relax today?"
 - Optional stress self-rating (0–3 scale).

Prediction & Insight

- Combine all data to predict **migraine probability within next 12 hours**.
- Display:
 - Risk level gauge (low/medium/high)
 - Key contributing factors (e.g. low sleep, pressure drop)
 - Personalized recommendations (e.g. hydrate, rest, avoid caffeine)

💬 Feedback & Personalization

- User logs feedback events:
 - `migraine_event`: {timestamp, pain_severity}
- Feedback: if migraine occurred, if prevention helped, if false positive.
- System retrains/adapts based on history (personal calibration).

🔔 Notifications & Engagement

- Notify user when:
 - High migraine risk detected.
 - Time to eat or hydrate (based on meal input or inactivity).
- Reminder prompts: "Did you drink coffee today?", "Have you relaxed today?"
- Integrate push notifications via FCM/APNS.

Relax Mode

- Dedicated "**Relax Mode**" page to reduce stress:
 - Guided breathing or short meditation sessions.
 - Ambient background music / haptic feedback.
- Track completion as an additional positive health input.

3. Data model additions

MigraineEvent:

```
{
  "user_id": "string",
  "timestamp": "datetime",
  "pain_severity": 1-10,
  "preventive_helped": true/false,
  "predicted": true/false
}
```

UserInputs:

```
{
  "caffeine": true/false,
  "meal_skipped": true/false,
  "self_reported_stress": 0-3,
  "relaxation_done": true/false
}
```

4. Feature flow summary

1. **Data Collection Layer**
2. Gather metrics via Health APIs + weather + calendar + user inputs.
3. **Processing & Storage Layer**
4. Send to FastAPI → stored in Firestore/Postgres → engineered features.
5. **Model Inference Layer**
6. Compute migraine risk and SHAP explanations.
7. **Output Layer**
8. Send probability + contributors + recommendations to mobile.
9. **Feedback Layer**
10. Log migraine events for retraining / personalization.

5. Feature plan & timeline

Pre-hackathon

- [] Repo setup (mobile, backend, model).
- [] Synthetic dataset generator including `migraine_event`.
- [] Baseline model (XGBoost / CatBoost) with SHAP analysis.
- [] Mock data flows from Health APIs & weather.
- [] Build static React Native UI screens for:
 - Dashboard (risk + contributors)
 - Input form (daily check-ins)
 - Relax Mode (breathing timer)
- [] FastAPI endpoints `/predict`, `/feedback`.

Hackathon Day 1

- [] Real-time mock data ingestion from app → backend.
- [] Risk calculation and explanation pipeline connected.
- [] Weather and calendar APIs integrated.

Hackathon Day 2

- [] Implement notification service (FCM/APNS) for risk & reminders.
- [] Enable user feedback logging and adaptive personalization.
- [] Finalize Relax Mode experience.

Hackathon Day 3

- [] Polish UX (animations, clarity of results).
 - [] Prepare demo scenarios + slide deck.
 - [] Rehearse 3–5 min pitch: *problem* → *solution* → *demo* → *impact*.
-

6. Success metrics

-  End-to-end demo: data → model → explanation → notification.
 - Clear interpretability: top 3 contributors displayed.
 -  Smooth UX with Relax Mode & meaningful notifications.
 -  Working feedback loop with personalization logic.
 -  Impact story: proactive prevention → improved life quality.
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7. Stretch ideas

- Real smartwatch integration (Fitbit, Apple Watch) for HRV.
 - On-device inference with TensorFlow Lite / CoreML.
 - LLM-based recommendation generator (“Your migraine risk is rising due to X and Y — consider...”).
 - Visual migraine timeline (calendar overlay).
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Summary:

AuraSense evolves into a holistic migraine prediction and prevention companion — integrating multi-source data (health, weather, calendar, user habits), producing interpretable 12-hour forecasts, delivering personalized notifications, and even guiding users through stress-reducing Relax Mode sessions. Ideal balance between scientific credibility, human empathy, and hackathon feasibility.